42nd Jim Isenberg Pacific Coast Gravity Meeting

UCR Center for Experimental Cosmology & Instrumentation

The 42nd Jim Isenberg Pacific Coast Gravity Meeting will be held on **Friday, February 27, 2026 - Saturday, February 28, 2026** at the University of California, Riverside.

In the tradition of the Pacific Coast Gravity Meetings, students and postdocs are strongly encouraged to participate, and all areas of gravitational physics—classical and quantum, theory and experiment—are welcome. We would like this meeting to serve as a communication medium among all branches of gravitational physics. Because this is a regional meeting, many participants will be from the Western U.S., but all are welcome.

The meeting is free to attend. There is **no registration fee**. However, we are not able to provide financial assistance to speakers and participants. A prize sponsored by the APS Division of Gravitational Physics (DGRAV) will be awarded for the best student talk.

Program

Below is the schedule of program events. All events will take place in person on the University of California, Riverside campus.

Thursday, February 26

5:00 PM - 6:00 PM **Welcome Reception**, University Lecture Hall (map)

6:00 PM - 7:00 PM **Frontiers of Cosmology Public Lecture**, University Lecture Hall (map)

Featuring APS President Brad Marston, with introductory remarks by 2017 Nobel Laureate Barry Barish

See the event page for more information.

Friday, February 27

8:30 AM - 9:00 AM **Coffee & Continental Breakfast**, HUB 302 North (map)

9:00 AM - 10:30 AM **Session I: Gravitational-Wave Data Science**, HUB 302 North (map)

Session Chair: Jonathan Richardson

Barry Barish, University of California, Riverside
Welcome and Overview

Ryan Johnson, California State University, Fullerton

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Including Gravitational Waveform Uncertainty in Bayesian Inference of Source Properties

Joseph Ribaudo, Embry-Riddle Aeronautical University
Semi-Classical Model of Detector Efficiency in the Context of Bipartite Gravitational-Wave Detection

Sophie Bini, California Institute of Technology
The impact of waveform systematics and Gaussian noise on the interpretation of GW231123

Tousif Islam, University of California, Santa Barbara
Binary black hole recoil kicks: modeling and astrophysical implications

Anjana Ashok, Oregon State University
Long-lived gravitational wave signals in current experiments

Siddharth Soni, University of California, Riverside
Machine Learning and Automation in LIGO Detector Characterization

10:30 AM - 11:00 AM **Coffee Break**

11:00 AM - 12:20 PM **Session II: Numerical Relativity**, HUB 302 North (map)
Session Chair: Jocelyn Read

Andrew Laeuger, California Institute of Technology
Probing Early Ringdown with the Direct Wave

Iago Mendes, California Institute of Technology
Numerical relativity for LISA sources: binary black hole initial data

Isabella Pretto, California Institute of Technology
Invariant identification of light rings in binary black hole simulations

Sepehr Salamat, University of Arizona

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Black Hole Meissner Effect and Singular Extremal Horizons

Vittoria Tommasini, California Institute of Technology
Data-Driven Acceleration of Eccentricity Reduction for Binary Black Hole Simulations

12:20 PM - 1:50 PM **Lunch Break**

1:50 PM - 3:20 PM **Session III: Gravity Theory**, HUB 302 North (map)
Session Chair: Preston Jones

Logan Finke, Embry-Riddle Aeroanutical University
Signatures of non-classicality in gravitational waves through bipartite detections

Quentin Bailey, Embry-Riddle Aeronautical University
Tests of spacetime symmetries using an effective field theory framework

Sawyer Star, Embry-Riddle Aeronautical University
Properties of Lorentz-symmetry breaking with a modified scalar field

Arunima Das, Arizona State University
Coherent and Squeezed States in Gravity

Colin Weller, California Institute of Technology
Isospectrality and Chiral Symmetry

Giuseppe Meluccio, Scuola Superiore Meridionale
Pre-geometric Gravity and Emergent Cosmology

Sam Gralla, University of Arizona
Can black holes evaporate past extremality?

3:20 PM - 3:50 PM **Coffee Break**

3:50 PM - 5:30 PM **Session IV: Gravitational-Wave Detector Science**, HUB 302 North (map)

Session Chair: Joshua Smith

Tyler Rosauer, University of California, Riverside
Next-Generation Wavefront Control Technology for Gravitational-Wave Detectors

Pooyan Goodarzi, University of California, Riverside
Impact of Beamsplitter Thermal Distortions on the Sensitivity of High-Power Gravitational-Wave Detectors

Kielo Hoops, California State University, Los Angeles
Characterization of Optical Scatterers in Coating Materials

Alina Sendowsky, California State University, Fullerton
Optical Scatter and Non-Contact Temperature Sensing of Cryogenically Cooled Crystalline Silicon

Maryam Ranjbar, University of California, Riverside
Automated Beam Repositioning for Mitigation of Nonuniform Absorption in Gravitational-Wave Detectors

Cynthia Liang, University of California, Riverside
Mitigating Point Absorber Risk in Gravitational-Wave Detectors

Michael Padilla, University of California, Riverside
Toroidal Mirrors for Enhanced Sensitivity in Current and Next-Generation Gravitational-Wave Detectors

Xuesi Ma, University of California, Riverside
Modeling and Characterizing Scattered Light Noise in the LIGO Detectors

5:30 PM - 7:30 PM **Networking Reception**, The Stable (map)
Light refreshments will be served, with drinks available for purchase.

Saturday, February 28

8:30 AM - 9:00 AM **Coffee & Continental Breakfast**, HUB 202 North (map)

8:30 AM - 9:00 AM **Coffee & Continental Breakfast**, HUB 302 North (map)

9:00 AM - 10:30 AM **Session V: Gravity Theory**, HUB 302 North (map)
Session Chair: Hai-Bo Yu

Dario Walter-Cardona, Embry-Riddle Aeronautical University
Systems of PDEs in Bumblebee Gravity

Ziyi Li, University of California, Davis
Aspects of Overspinning and Extremal Defects in AdS_3

Anajak Phonchantuek, California State University, Fresno
Nonlinear chiral forms in Sen formulation of superstring theory

Eduardo Guendelman, Ben-Gurion University
Dark Energy and Dark Matter with Dynamical Tension Strings

10:30 AM - 11:00 AM **Coffee Break**

11:00 AM - 12:20 PM **Session VI: Numerical Relativity**, HUB 302 North (map)
Session Chair: Geoffrey Lovelace

Dongze Sun, California Institute of Technology
Parameter matching between PN and NR via an inner-zone matching for binary black holes

Guido Da Re, California Institute of Technology
Initial data construction for Cauchy-characteristic evolution

Alexandra Macedo, California State University, Fullerton
Writing Tensor Equations in SpECTRE

Jack Kutcka, California State University, Fullerton
Testing SpECTRE's Outer Boundary Condition and Stability

Taylor Knapp, California Institute of Technology
Full control of the parameters of eccentric and precessing SpEC waveforms

Jose Pineda, California State University, Fullerton

Comparing SpEC and SpECTRE evolutions of the same binary-black-hole initial data

12:20 PM - 1:50 PM **Lunch Break**

1:50 PM - 3:20 PM **Session VII: Future Experiments**, HUB 302 North (map)
Session Chair: Siddharth Soni

Amber Romero, California State University, Fullerton
Improving techniques for identifying and evaluating sites for Cosmic Explorer

Cecilia Ochoa, University of California, Riverside
Mirror Design for Optimal Higher Order Mode Placement in Cosmic Explorer

Briana Chen, California Institute of Technology
Using quantum metrology to advance fundamental physics

Mark Nguyen, Northwestern University
Precision Tests of Gravity with Optically Levitated Nanoparticles

Sujoy Modak, California State Polytechnic University, Pomona
A Pathway for Probing Weak Quantum-Gravity Deformations Using Present-Day Technology

Spencer Johnson, Jet Propulsion Laboratory
Toward Gravitational Redshift Measurements with Quantum States of Light

Douglas Singleton, California State University, Fresno
Temporal Pound-Rebka experiment as gravitational Aharonov-Bohm effect

3:20 PM - 3:50 PM **Coffee Break**

3:50 PM - 5:20 PM **Session VIII: Gravity Theory**, HUB 302 North (map)
Session Chair: Douglas Singleton

Daniel Hooker, California State University, Fresno

Symmetrized operators versus modified integration measure in Generalized Uncertainty Principle models

Zach Ghosoph, Clovis Community College
From Linearized GR to Quantum Gravity via Canonical Quantization

Diego Hidalgo, Clovis Community College
From Quantum Gravity to Linearized General Relativity via Bohr Correspondence Principle

Rose Zanoni, Clovis Community College
Resolving the Cosmological Constant Problem via quantum space-time uncertainty

Nathan Inan, California State University, Fresno
Evolution of Quantum Space-Time Uncertainty on Cosmological Scales

Michael Bishop, California State University, Fresno
Minimal lengths in 3D via the Generalized Uncertainty Principle

5:20 PM - 5:30 PM **Announcement of DGRAV Student Prize Winner**, HUB 302 North (map)